

Chapter 11, Problem 1P

Problem

If $v(t) = 160 \cos 50t$ V and $i(t) = -33 \sin(50t - 30^\circ)$ A, calculate the instantaneous power and the average power.

Step-by-step solution

Step 1 of 3

Consider the voltage $v(t)$ as,

$$v(t) = 160 \cos(50t) \text{ V}.$$

Consider the current $i(t)$ as,

$$\begin{aligned} i(t) &= -33 \sin(50t - 30^\circ) \text{ A} \\ &= 33 \sin(50t - 30^\circ + 180^\circ - 90^\circ) \text{ A} \\ &= 33 \cos(50t + 60^\circ) \text{ A} \end{aligned}$$

Calculate the instantaneous power $p(t)$.

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= (160 \cos(50t) \text{ V})[33 \cos(50t + 60^\circ) \text{ A}] \\ &= 5280 \cos(50t) \cos(50t + 60^\circ) \end{aligned}$$

Step 2 of 3

Apply the following trigonometry identity to the equation.

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

The following is the instantaneous power p :

$$\begin{aligned} p(t) &= \frac{5280}{2} [\cos(50t + 50t + 60^\circ) + \cos(50t - 50t - 60^\circ)] \\ &= 2640 [\cos(100t + 60^\circ) + \cos(-60^\circ)] \\ &= 2640 [\cos(100t + 60^\circ) + 0.5] \\ &= 1.320 + 2.640 \cos(100t + 60^\circ) \text{ kW} \end{aligned}$$

Therefore, the value of instantaneous power $p(t)$ is $\boxed{1.320 + 2.640 \cos(100t + 60^\circ) \text{ kW}}$.

Step 3 of 3

Calculate the average power P .

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Replace 160 for V_m , 33 for I_m , 0° for θ_v , and 60° for θ_i in equation.

$$\begin{aligned} P &= \frac{1}{2} (160)(33) \cos(0^\circ - 60^\circ) \\ &= \frac{5280}{2} (0.5) \\ &= 1.320 \text{ kW} \end{aligned}$$

Therefore, the average power P is 1.320 kW.